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Determination of optimal rotation period under stochastic wood and carbon prices

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Abstract

The valuation of forest stands is traditionally based on a profit calculus involving revenue from wood sales and associated costs. Currently, the role of carbon management in forests is actively discussed. In a stochastic setting we extend the analysis of the optimal rotation period by considering uncertain revenue streams from carbon trading. We develop a real options model given uncertainties in future wood and CO_2 price behaviour. A detailed sensitivity analysis of the numerical results for both cases – with and without carbon sequestration – is provided. We find that optimal rotation periods vary considerably with (i) the type of price process, (ii) the way how carbon income is defined, and (iii) the selection of discount rates.
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1. Introduction

The controversy of optimal forest management has a long history in forestry economics. The proposal, in 1849, by Faustmann to consider discounting in a dynamic setting has substantially impacted the thinking about optimal forest management (Faustmann, 1849). In particular, in central Europe the Faustmann model has been raising debates since its inception and discussions on a more international level continue to be lively (see e.g. Brazee, 2001; Newman, 2002).

During the last three decades many authors attempted to extend the tree-cutting problem, in order to incorporate stochastic behaviour. The first studies within such context has probably been papers by Lembersky and Johnson (1975) and Brock et al. (1982). In Lembersky and Johnson (1975) authors deal with the stand management problem under price and stand response uncertainty in infinite time horizon. The uncertainties are modelled using state transition probabilities in discrete space. Since Lembersky and Johnson consider only a finite number of states and management actions, the optimal policy can be expressed as a solution

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to a specific functional equation and thus obtained iteratively. On the other hand, Brock et al. consider the harvesting problem in a more general context: the context of stochastic capital theory. They model the tree size using both discrete and continuous processes and prove that the so-called *barrier strategy* is optimal. However, their model setup is only a simple one: the value of the tree they consider is given merely by its size. In a subsequent paper Miller and Voltaire (1983) extended the work by Brock et al. (1982) to the analysis of the ongoing rotation problem.

As Newman (2002) summarizes, since the late 1980s researchers' interest in the optimal harvesting problem under uncertainty has been increasing rapidly. The stylized problems considering different type of uncertainties have been developed in an attempt to extend the Faustmann's formula for ongoing rotations on one hand and the so-called Wicksellian tree-cutting problem (that is, the single rotation setup) on the other hand, in order to handle stochastic behaviour. For example, the papers by Brazee and Mendelsohn (1988), Haight and Holmes (1991), and Clarke and Reed (1989), belong to the most significant contributions to the field of the optimal harvesting problem and were a foundation for many other publications that appeared in the last decade.

In particular, Brazee and Mendelsohn analyze an impact of price uncertainty within Faustmann's framework, assuming that clear-cut is the only management decision. Haight and Holmes pointed out the importance of the specification of the underlying timber price model. Using the dynamic programming formulation in discrete space they have determined the effect of different empirically tested timber price models on the optimal harvest policy. Clarke and Reed address the optimal harvesting problem in the continuous-time. Assuming the price and growth uncertainty (both modelled by a geometric Brownian motion) they have derived the harvesting rules for single rotations and expressed the optimal cutting rule as a barrier.

Even in the first studies on the optimal rotation under uncertainty (cf. Brock et al., 1982) many authors have pointed out that the problem of determining the single optimal rotation (or, equally, the Wicksellian tree-cutting problem) is actually a sort of optimal stopping problem known from various economic areas. In corporate finance theory the capital budgeting problems of this type (i.e. the problems related to uncertainty for which it is important to determine management's strategic reaction to the changing conditions) are nowadays commonly analyzed within the real options framework (for more details see, e.g., Dixit and Pindyck, 1994). The papers by Morck et al. (1989) and Thomson (1992) are probably among the first applications in the forestry, which have adopted the real options approach and employed the solution techniques known from the financial options theory. Thomson has proposed the model in a discrete framework and applied the lattice method. Morck et al. on the other hand have calculated the value of forestry lease under stochastic timber price and stochastic timber inventory using the contingent claim approach (cf. Merton, 1973).

More recently, a number of studies applying the real options methodology to forest management problems have been published. Their application areas differ mostly in the model formulation, in particular the way how uncertainty is treated. Commonly, the wood price or forest growth, or even both, are assumed to be stochastic (Insley, 2002; Saphores, 2003; Alvarez and Koskela, 2004b). In addition, Alvarez and Koskela (2004a) have analyzed optimal forest management under stochastic interest rate. Nonetheless, these studies provide different types of results: either a numerical or theoretical analysis of the problem is presented. In particular, Insley concentrates on the numerical illustration (when the tree harvesting problem is formulated as a linear complementarity problem), while Saphores, Alvarez and Koskela present mainly the theoretical deductions characterizing the optimal harvesting policy. The question of the proper specification of the underlying uncertainty has also been addressed: the timber price development is either described by the geometric Brownian motion or a mean-reverting process, the latter being more suitable for analyzing empirical data (see Insley, 2002).

Except the real options applications there appeared other approaches related to the stochastic optimal rotation models. For instance, Willassen (1998) applied impulse control method to the problem of generalization of the Faustmann's formula to the case of stochastic forest growth, and Lu and Gong (2003) determined the optimal thinning and harvest strategy by optimizing the coefficient of the optimal stocking level function. Despite this fact, we believe that the shift from the Faustmann's traditional net present value (NPV) approach to the real options framework has been a major innovation

in the forest management science. However, the valuation of joint production of ancillary services and wood products has not yet been sufficiently appraised in the scientific literature. In particular, the introduction of carbon trading might lead to a change in the investment environment of forest owners. There is a need for decision support tools advising forest managers to operate in this new and uncertain business environment.

According to the Kyoto Protocol mainly developed countries (Annex I countries) are contracted to reduce their greenhouse gas (GHG) emissions. It is foreseen that market based mechanisms shall help countries meeting their emissions commitments. These flexibility mechanisms assume the use of sinks and carbon trading.[1] Carbon trading might be realized as follows: Depending on the country specific rules, the CO_2 emitters receive a certain amount of carbon credits. If the amount of the released CO_2 exceeds the assigned amount, the owner can either try to reduce the amount of CO_2 by installing a new, more environmentally friendly technology or, in some cases, it might temporarily be more profitable to purchase additional carbon credits. A forest owner could potentially provide carbon credits filling this supply gap (see, e.g. Binkley et al., 2002).

A forest is recognized as an important carbon sink: growing trees store a significant amount of CO_2 during growth. The basic principles of how to treat sinks were laid down at the Marrakesh Accords of the 8th Session of the Conference of the Parties of the United Nations Framework Convention of Climate Change. The exact role of the forest in the carbon trading mechanism is, however, still an open question. A basic specification of the carbon trading scheme requires us to establish the rules of reward for carbon sequestration, that is the frequency and the form of carbon payments must be determined (e.g., annual payments for each additional increment or only one payment at the beginning of the forest plantation). Another crucial assumption is setting the forest owner's obligations at the time of harvest, since once the forest is harvested, much of the stored carbon is converted back to CO_2 again. In general, there is no definite conclusion from the debate on this liability issue: hence, a specific carbon scheme can establish a proportion of the released CO_2 for which the forest owner is obliged to provide carbon credits after harvesting has been performed. A more detailed discussion will be presented in Section 3.

Over the last ten years, a number of studies dealing with carbon sequestration and the impact on the optimal forest management have been published. However, most of these studies analyze the impact of carbon trading using NPV analysis (see, e.g., Romero et al., 1998; Gong and Kriström, 1999; Benitez and Obersteiner, 2006). There are only a few papers which examine the impact of carbon sequestration on the optimal rotation period within a real options framework. Among them, a paper by Guthrie and Kumareswaran (2003) is closest to our study. The authors introduce three different carbon allocation regimes (one regime allows even deforestation). In contrast to the work presented in this paper the price of carbon credits is kept constant in their analysis. We will try to augment current analysis on additional dimension of complexity and establish a model of *two stochastic sources* of revenue allowing us to exploit the independence of the stochastic price processes of both the market for wood and the market for carbon. Thus, the main aim of this study is to develop a model for the optimal forest management decisions under the presence of carbon and wood sales payment streams. This approach is related to multifactor options pricing models, which are usually very difficult to handle. In our setting the real options methodology is used to determine the optimal rotation period under the assumption of stochastic wood *and* stochastic carbon prices. A detailed sensitivity analysis of the numerical results for including and excluding carbon sequestration is provided.

The paper is organized as follows. A real options model, including and excluding carbon sequestration under stochastic wood and carbon prices is introduced in Section 2. In Section 3 we describe the data used to calibrate the model, and propose three carbon schemes, which might be applied in the forestry sector. Section 4 contains a comprehensive collection of the numerical results and a detailed sensitivity analysis of the model parameters is provided. Finally, Section 5 discusses the implications of our results together with possible future model extensions.

2. The model

The aim of this modelling exercise is to analyze optimal forest management decisions when carbon

[1] A detailed discussion of this topic can be found e.g. in Sedjo, 2001.

payments are present. For reasons of clarity to present the model we deal exclusively with the problem of determining the single optimal rotation period. The set of modelled management options consists hence primarily of the following two actions: one can either postpone the harvest and continue growing the forest, or, alternatively, perform the harvest and sell the land.

We formulate this problem from the forest owner's point of view. In order to cope with complexity, we first present the simplified model where no carbon payments are considered.

2.1. The model without carbon sequestration

In this case, the value of the forest is determined by two main quantities: the wood price P and the forest stand volume S. We assume that wood prices build a time series that follows the mean-reverting process:

$$\mathrm{d}P_t = \alpha(\mu - \ln P_t)P_t dt + \sigma P_t \mathrm{d}W_t. \tag{1}$$

In this model, the price P reverts to the long-term level $\overline{P} = \exp(\mu)$ at a speed given by the mean reversion rate, α. The term $\mathrm{d}W$ denotes the increment of a standard Wiener process and σ is a volatility of the process. Note, that the uncertainty in wood prices modeled by Eq. (1) meets the Markov property: given the price P at time t the values of P before time t are irrelevant when predicting the future behaviour of price development.

Furthermore, we assume that if left unattended the forest stand evolves according to the specific deterministic function g. This evolution can be described by:

$$\mathrm{d}S_t = g(S_t)dt. \tag{2}$$

Under these assumptions, the forest owner's maximization problem, that is the problem to determine the optimal management strategies $\{a(t)\}_{t=1}^{T}$ that should be performed within the fixed time horizon T, can be formulated in terms of maximization of a so-called *value function* V:

$$V(t, S_t, P_t) = \max_{i \in \mathcal{A}(t,S_t,P_t)} \{\pi_i(S_t, P_t) + e^{-\beta \Delta t} \mathbb{E}(V(t + \Delta t, S_{t+\Delta t}, P_{t+\Delta t}) | S_t, P_t)\}. \tag{3}$$

The relation (3) is the well-known Bellmann equation (see, e.g., Dixit and Pindyck, 1994) and can be explained as follows: Under the optimality principle, the value of function V at time t can be decomposed into the sum of the immediate profit π_i and the expected discounted continuation value $E(V)$ (the discount rate is denoted by β). Now, denote the index i in Eq. (3), which corresponds to the optimal decision at time t, by $a(t)$. Then $a(t)$ is the local decision (or, equally, the "action") which maximizes that sum. To be more strict, the possible actions i should be chosen only among current feasible decisions described by the set $\mathcal{A}(t, S_t, P_t)$.

As we have already discussed in the beginning of this section our analysis only deals with the following two actions:

- $i=1$: postpone the harvest, and
- $i=2$: harvest and sell the land.

We assume, that there are no restrictions on the feasibility of these actions, that is $\mathcal{A}(t, S_t, P_t) = \{1, 2\}$ for any t, S_t, and P_t.

However, the model is proposed so that it can easily be extended to cover a broader class of forest management options. For instance, an option to perform thinning or replantation (multiple periods) can be included. We will, therefore, continue presenting the model along this general action setup.

In order to prevent speculations at the real estate market from biasing our results, we assume the price of land to be zero. In case of no carbon trading, the immediate profit π_i is generated only by selling wood, that is:

$$\pi_i(S_t, P_t) = f_i(S_t)(P_t - C_i). \tag{4}$$

Here, $f_i(S_t)$ denotes the amount of wood harvested if action i is performed. If the harvest is performed ($i=2$), then $f_2(S_t) = S_t$ (that is, all trees are cut). On the other hand, if the harvest is postponed ($i=1$), $f_1(S_t) = 0$. The value C_i remains the same for the costs per m^3 associated with performing action i (for instance, the costs of harvesting). For simplicity, we set costs to be constant. Once again, the function π_i is proposed so that the thinning option could easily be included. In such a case, $f_i(S_t)$ would stand for the volume of wood harvested due to thinning, and C_i would represent the costs associated with thinning itself, which generally differ from the harvesting costs.

2.1.1. Decision points

We assume that management decisions can be implemented only at fixed time points $t_0, t_1, \ldots, t_n = T$, e.g., only on a yearly basis. Furthermore, we assume that there is no delay between the realization of the respective action and its related payoff. This means that action payments are also received at these specified time points only.

Under these assumptions, during the period when no decision can be adopted, i.e., for $t \in (t_j, t_{j+1}), j = 0, \ldots, n$, the immediate profit is zero.

However, the forest grows continuously between these decision points. Moreover, wood prices might also change within this period. Hence, the Bellman Eq. (3) for the value function is modified to:

$$\text{for } t \in (t_j, t_{j+1}): \; V(t, S_t, P_t) = 0 + e^{-\beta \Delta t} \mathbb{E}(V(t + \Delta t, S_{t+\Delta t}, P_{t+\Delta t}) | S_t, P_t). \tag{5}$$

Now we apply the steps described in Dixit and Pindyck's (1994) book. We multiply Eq. (5) by $e^{\beta \Delta t}$ and rearrange this as follows:

$$\begin{aligned}(e^{\beta \Delta t} - 1) V(t, S_t, P_t) &= \mathbb{E}(V(t + \Delta t, S_{t+\Delta t}, P_{t+\Delta t}) \\ &\quad | S_t, P_t) - V(t, S_t, P_t) \\ &= \mathbb{E}(\Delta V | S_t, P_t).\end{aligned}$$

That is,

$$\beta V(t, S_t, P_t) = \beta \frac{1}{(e^{\beta \Delta t} - 1)} \mathbb{E}(\Delta V(t, S_t, P_t) | S_t, P_t)$$

Expanding $e^{\beta \Delta t}$ into the Taylor series and letting Δt tend to zero we obtain

$$\beta V(t, S_t, P_t) = \frac{1}{\mathrm{d}t} E(\mathrm{d}V(t, S_t, P_t) | S_t, P_t)$$

As $\mathbb{E}(\mathrm{d}W) = 0$, applying the Itô lemma we achieve:

$$\begin{aligned}\beta V(t, S_t, P_t) = &\frac{\partial V(t, S_t, P_t)}{\partial t} \\ &+ \frac{\partial V(t, S_t, P_t)}{\partial S_t} g(S_t) \\ &+ \alpha(\mu - \ln P_t) P_t \frac{\partial V(t, S_t, P_t)}{\partial P_t} \\ &+ \frac{1}{2} \sigma^2 P_t^2 \frac{\partial V^2(t, S_t, P_t)}{\partial P_t^2}.\end{aligned} \tag{6}$$

A solution of the partial differential Eq. (6) enables us to describe the value of the function $V(t, S_t, P_t)$ between the decision points.

In order to solve the proposed stochastic dynamic problem it is necessary to specify the boundary conditions. First, we constrain the terminal action to be $a(T) = 2$, i.e., the "harvest and sell" option must be adopted at the final decision point T or earlier. This condition implies that the value of function V at time T must be:

$$V(T, S_T, P_T) = \pi_2(S_T, P_T).$$

Further, the following conditions must hold:

- $V(t, 0, P_t) = 0$,
- $V(t_j, S_{t_j}, P_{t_j}) = \pi_{a(t_j)} + \lim_{t \to t_j} + V(t, S_t - f_{a(t_j)}(S_{t_j}), P_t)$, if $a(t_j) \neq 2$, and
- $V(t_j, S_{t_j}, P_{t_j}) = \pi_2(S_{t_j}, P_{t_j})$, if $a(t_j) = 2$.

The first condition reflects the fact that if there is currently no standing volume ($S_t = 0$), then S_t stays zero forever, that is no future afforestation is assumed. Moreover, when an option is exercised, the value of function V changes. This change can be described by the latter two conditions.

2.2. The model with carbon sequestration

In this section we extend the model described in Section 2.1 so that the payments arising from carbon sequestration will be considered.

In addition to the assumptions proposed in the previous section, suppose that the CO_2 prices can be modelled by the geometric Brownian motion (GBM) process with instantaneous drift μ^c and volatility σ^c:

$$\mathrm{d}P_t^c = \mu^c P_t^c \mathrm{d}t + \sigma^c P_t^c \mathrm{d}W_t^c. \tag{7}$$

Denote the correlation between the increment of the Wiener processes $\mathrm{d}W$ and $\mathrm{d}W^c$ as ρ.

A simple modification of the Bellman Eq. (3) with respect to carbon sequestration leads to:

$$\begin{aligned}V(t, S_t, P_t, P_t^c) = &\max_{i \in \mathcal{A}(t, S_t, P_t, P_t^c)} \Big\{ \pi_i^c(S_t, P_t, P_t^c) \\ &+ e^{-\beta \Delta t} \mathbb{E}(V(t + \Delta t, S_{t+\Delta t}, P_{t+\Delta t}, P_{t+\Delta t}^c) | S_t, P_t, P_t^c) \Big\}.\end{aligned} \tag{8}$$

This time, the immediate profit flow π_i^c consists of two parts:

$$\pi_i^c(S_t, P_t, P_t^c) = f_i(S_t)(P_t - C_i) + \gamma \Delta S_t P_t^c - f_i^c(S_t) P_t^c. \quad (9)$$

The first term of this statement coincides with the profit described by Eq. (4). The second term includes the profit arising from carbon sequestration. Here we assume that carbon payments are received for any increment in standing volume (see a discussion in Section 3). The parameter γ is a conversion factor and states how many tons of CO_2 are sequestered in 1 m^3 of wood (i.e., $\gamma \approx 0.25 \times 44/12$). Once the harvest is performed the appropriate (depending on the carbon scheme) amount of carbon credits must be purchased back. The last term in statement (9) can be expanded as:

$$f_i^c(S_t) = \theta \gamma f_i(S_t),$$

where θ is the proportion of total CO_2 uptake in the harvested wood for which carbon credits must be purchased (see Section 3 for more details). In particular,

$$f_1^c(S_t) = 0$$

$$f_2^c(S_t) = \theta \gamma S_t.$$

Using the same steps as in Section 2.1, we derive the following partial differential equation:[2]

$$\begin{aligned} \beta V(t, S_t, P_t, P_t^c) = {} & \gamma g(S_t) P_t^c + \frac{\partial V}{\partial t} + \frac{\partial V}{\partial S_t} g(S_t) \\ & + \alpha(\mu - \ln P_t) P_t \frac{\partial V}{\partial P_t} + \mu^c P_t^c \frac{\partial V}{\partial P_t^c} \\ & + \frac{1}{2}\left(\sigma^2 P_t^2 \frac{\partial^2 V}{\partial P_t^2} + (\sigma^c P_t^c)^2 \frac{\partial^2 V}{\partial (P_t^c)^2}\right) \\ & + \rho \sigma \sigma^c P_t P_t^c \frac{\partial^2 V}{\partial P_t \partial P_t^c}, \quad t \in (t_j, t_{j+1}) \end{aligned} \quad (10)$$

that is valid for the period between decision points t_j.

3. Inputs

In the numerical analysis we perform a case study for an even-aged forest which could be located in Tirol, Austria on an area of one hectare. We chose spruce as the representative species as it is the most common one in this region.

There are three major inputs to our model: wood price, carbon credit price (CO_2 price) and growth function, which are necessary to calibrate the data for this specific case. Moreover, the cost of harvesting must be determined: we set $C_2 = 30$ €/m^3 based on information of the regional estimates from the Austrian Energy Agency. Finally, the carbon crediting scheme θ should be discussed and specified.

3.1. Wood price

For the purposes of our Austrian case study, we used the historical time series for the Tirol region collected by DVT-Daten-Verarbeitung-Tirol GmbH. The sample in question consists of the monthly prices for spruce for the period starting January 1994 to December 2004. Prices are quoted in €/m^3.

Fig. 1 describes the development of spruce prices. The price level decreases in the middle of 2000 is most likely caused by the wind storm in June 2000 in Austria.[3] The average price for the period in question is 76 €/m^3 and its standard deviation is 6.74.

Recall that we model wood prices as the mean-reverting process (1). Denote $p_t = \ln P_t$. Then process (1) can be rewritten as:

$$dp_t = \alpha(\bar{p} - p_t)dt + \sigma dW_t, \quad (11)$$

where $\bar{p} = \mu - \sigma^2/2\alpha$. The process (11) is known as the arithmetic Ornstein–Uhlenbeck process. The log-form of Eq. (1) enables us to estimate the parameters of that process using the standard econometric techniques (see, e.g., Dixit and Pindyck, 1994 for details). After a few simple calculations we obtain the following values for the original wood price process (1): $\hat{\alpha} = 0.4556$, $\hat{\mu} = 4.313$, $\hat{\sigma} = 0.0924$.

3.2. CO_2 price

In the proposed model, the CO_2 price is an exogenous input. The carbon market does not exist

[2] Actually, the multivariate version of the Itô lemma must be applied. For reference see, e.g., Björk, 1998.

[3] Source: Emergency Events Database, http://www.em-dat.net/disasters.

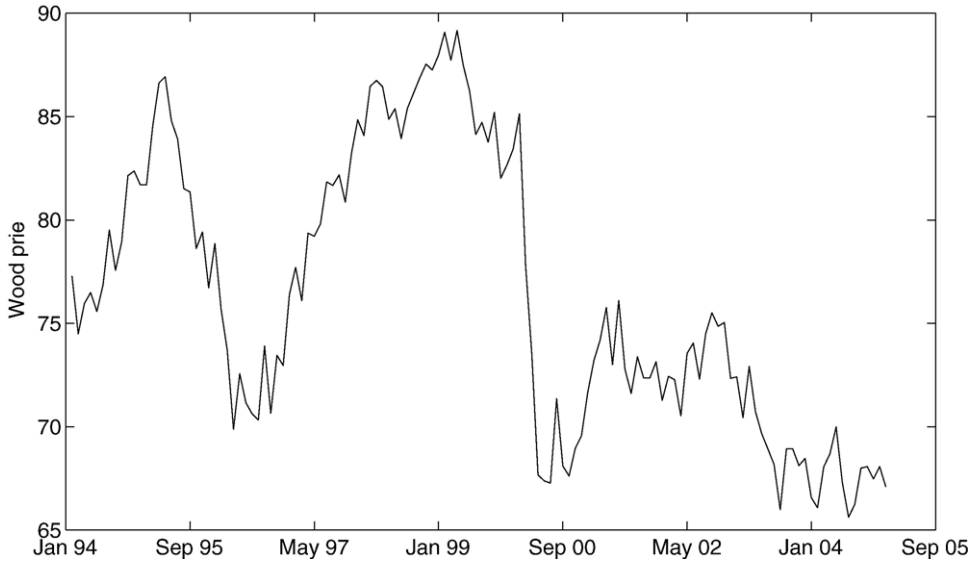

Fig. 1. Spruce price development during the period January 1994–December 2004 in Tirol, Austria. Prices are quoted in €/m^3.

long enough to provide any reliable information about price development. Nevertheless, several global models dealing with this issue exist and yield relevant information. As the basis for our estimation of CO_2 price development we consider the carbon values calculated by the IIASA MESSAGE model.[4] In this case, the carbon value can be interpreted either as a carbon tax or a value of an emission permit created for a carbon trading market in a carbon emission constrained world aiming to meet a greenhouse gas stabilization target.

In the stabilization scenarios used for this exercise, CO_2 prices grow steadily from about 5 €/tCO_2 in 2010 to about 7–17 €/tCO_2 in 2050, then to about 33–70 €/tCO_2 in 2070 and, finally, to about 110–136 €/tCO_2 in 2100.

In order to constrain the range within which prices are allowed to fluctuate, we have selected the A2–550t price trajectory as an approximation of the upper bound and the B2–550t price trajectory as an approximation of the lower bound of the CO_2 price range. In Fig. 2 scenarios A2 and B2 are depicted.

In the stochastic model described above, we assume that CO_2 prices follow a geometric Brownian motion (7). We assume (under subjective consideration) that the correlation ρ between dW in Eq. (1) and dW^c in Eq. (7) is 0.5. Regarding the CO_2 prices calculated using the MESSAGE scenarios, we fix the parameters of the process (7) to $\mu^c = 0.03625$ and $\sigma^c = 0.16604$.

3.3. Growth function

According to Birch (1999) we model the growth of the forest stand $g(S_t)$ as follows:

$$g(S_t) = \dot{S}_t = S_t(A_1 + A_2 S_t^{x-1}) \tag{12}$$

The parameters of the growth curve (12) follow on from the standing volume data contained in the Marschall yield table (see Marschall, 1975). In this study we deal with the most typical Austrian species *Spruce Bayern* and assume a site class 11. The estimated values for this case are $\hat{A}_1 = -0.095518$, $\hat{A}_2 = 0.81215$, and $\hat{x} = 0.6823$. The maximum achievable volume on the area of one hectare is 843 m^3.

3.4. Carbon crediting schemes

Growing trees capture CO_2 from the atmosphere. Most of that carbon remains stored *in situ* until harvesting operations remove carbon from the site. At the time of harvest, the stored carbon is released into the atmosphere, if the wood is used for bioenergy purposes, otherwise the embodied carbon enters a wood product

[4] A detailed description can be found in Riahi and Roehrl, 2001.

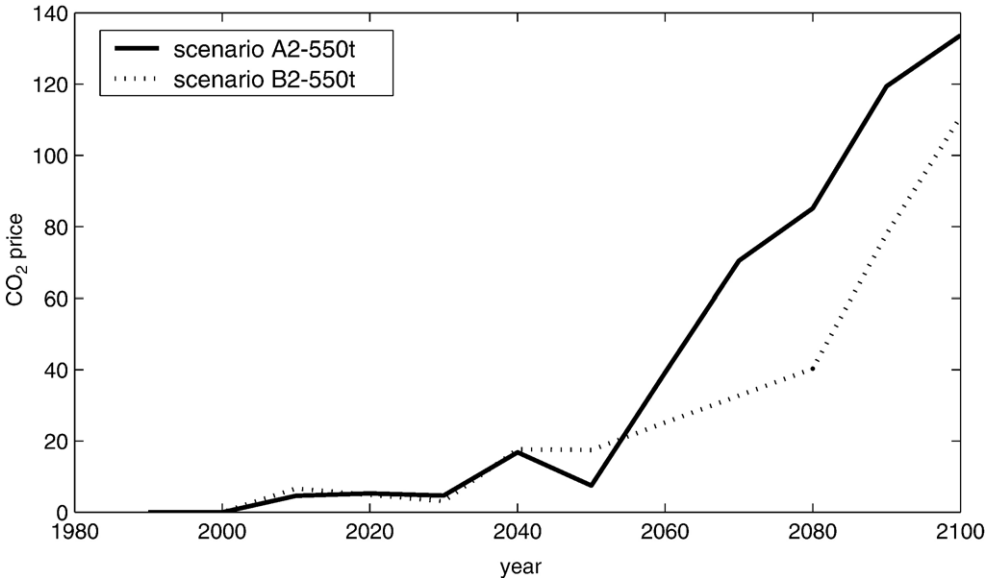

Fig. 2. CO_2 prices calculated on the basis of the MESSAGE scenarios A2 and B2. Prices are quoted in €/tCO_2.

pool. Our simple carbon scheme is built on these principles. We assume that during the growing phase of the forest, the owner is allowed to *sell* carbon credits for the annual increase in standing biomass. Thus, the benefits are paid annually. At harvest time the forest owner is obligated to buy carbon credits accounting for the carbon loss due to the removal of wood.

Due to uncertainties on carbon accounting principles it is still an open issue how many carbon credits the owner must buy at harvest time. Carbon crediting crucially depends on how liabilities for carbon losses or removals are treated. Recall in Section 2.2 by θ we have denoted the *proportion of total carbon* embodied in harvested wood for which credits must be purchased. In our numerical analysis (see Section 4) we consider the following three carbon crediting schemes:[5]

- $\theta=1$: In this case, we assume full liability for the embodied carbon at the end of the rotation period. Implicitly, it is assumed that after harvesting 100% of the carbon is released immediately into the atmosphere.
- $\theta=0.7$: In this case, there is partial liability for the embodied carbon corresponding to a situation where the carbon enters a pool of decaying wood or the wood harvested is burned in a combined heat and power plant after transportation and the net atmospheric benefit due to fossil fuel substitution is 30%.
- $\theta=0.3$: In this case, there is low liability for the embodied carbon corresponding to a case where the harvested wood is used in a bioenergy plant with subsequent carbon capture and sequestration (BECS). The 30% reflects the inefficiencies of the process and energy requirements for transportation. This case could, however, equally correspond to a situation where large amounts of wood enter a slowly decaying wood or wood products pool.

4. Results and discussion

In order to characterize the forest owner's behaviour, we have performed a number of numerical experiments. Our main objective has been to analyze the optimal forest management decisions under various parameter setups. We have mainly focused on the following two issues. First, we explored the sensitivity of the length of the optimal rotation period to the discount rate. We studied this problem under different carbon crediting schemes (see Section 3). Furthermore, we tried to understand how price uncertainty influences the optimal decisions. We

[5] A similar setup has been introduced by Appels (2001).

compared numerical experiments mainly on the basis of the resulting single optimal rotation period length.

Solving the problem formulated in Section 2.1 or in 2.2, respectively, requires us to implement a dynamic programming algorithm moving backwards with respect to decision time points. Moreover, for t between the decision points, the partial differential Eqs. (6) and (10), respectively, must be solved. In all our numerical experiments the maximum length of the rotation period T has been set to 200 years. Once the optimal decisions are calculated we may simulate the prices (P, P^c) over the interval 0…, T using properties of the processes (1) and (7), respectively. Starting with a 15 year old forest,[6] for each simulation we can determine the optimal rotation period, i.e., the optimal moment to perform the "harvest and sell" action.

4.1. The benchmark case (no carbon sequestration)

The main aim of this paper is to analyze the change in optimal forest management when carbon sequestration is additionally considered. In order to study this effect we first analyze the case without carbon trading as proposed in Section 2.1. In addition, we simplify the model a bit more, as we assume wood prices to be constant. Later on, we refer to this benchmark case and compare the length of the optimal rotation period with those resulting from the more sophisticated models.

If the wood price P_t is assumed to stay constant, the optimization problem defined by Eq. (3) can be solved analytically. It can easily be derived that to perform the "harvest and sell" action is optimal if, and only if, the following condition for the forest stand volume and the discount rate holds:

$$\frac{g(S_T)}{S_T} = \beta. \tag{13}$$

As we deal with even-aged forest only, we can apply the 1-1 correspondence between the standing volume and the forest age and obtain the optimal rotation period. For the standard values of the discount rate applied in the forestry sector (1%, 2%, and 3%) the optimal rotation lengths are 80, 65, and 51 years, respectively.

4.2. Deterministic setup with carbon sequestration

In this section we explore the model incorporating carbon sequestration, as described in Section 2.2. The following sections present detailed analysis of the deterministic and stochastic setup. We provide the results showing key differences between these two approaches.

4.2.1. Constant wood price, constant CO_2 price

Again, we start our analysis with the simplest case: we keep the wood price as well as the CO_2 price constant, that is we set $\alpha = 0$, $\sigma = 0$, $\mu^c = 0$, and $\sigma^c = 0$.

If a carbon scheme is present, the forest owner faces a more complex problem: he/she must determine the optimal time to perform the harvest not only with respect to the profit gained from selling wood, but also he/she must take into account the payoffs generated by carbon sequestration. Given the carbon scheme proposed in Section 3, the possibility of selling carbon credits during the growing phase of the forest leads to an increase of its value. Hence, extending the rotation period is favorable not only because more wood volume can be sold, but it also means that more carbon credits can be sold as well. Moreover, postponing the harvest also means that carbon credits, which are necessary to cover CO_2 release at harvest time, must be purchased back later (in other words, due to discounting there is a difference in the time-value of carbon payments). It is important to notice here that under $\theta = 1$ carbon trading is beneficial for the forest owner only because of this reason. On the contrary, discounting also acts against extending the rotation period: the longer the owner waits, the later he/she receives the profit determined by selling wood. Hence, when carbon trading occurs finding the optimal action $a(t)$ requires us to handle the tradeoff relation between selling wood and carbon sequestration.

Now, we compare this deduction with the results of the numerical computations. Table 1 presents the optimal rotation period under the different parameter setups. In addition to the sensitivity analysis for the discount rate (1%, 2% and 3%, respectively), we provide the results valid for all three proposed carbon schemes (i.e., we set $\theta = 1$, 0.7 and 0.3, respectively).

[6] 15 years was chosen as starting value because of the data used for growth curve modelling; namely, no standing volume data for younger trees are presented in the yield tables. Actually, it would be tedious and unreliable to collect the growth data for younger forests and therefore, our considerations concern only forests that are older than 15 years.

Table 1
Optimal rotation period (in years) as a function of carbon scheme, discount rate and under the assumption that the *constant* wood price is 76 €/m^3 and the *constant* CO_2 price is 10 and 50 €/tCO_2, respectively

Carbon scheme θ	Discount rate β	Optimal rotation $P^c=10$	Optimal rotation $P^c=50$
1	1%	86	155
1	2%	68	127
1	3%	56	118
0.7	1%	86	118
0.7	2%	68	96
0.7	3%	56	86
0.3	1%	86	101
0.3	2%	68	86
0.3	3%	56	72

Moreover, we consider two significantly different values for the CO_2 price: 10 €/tCO_2 and 50 €/tCO_2.[7] The value of the wood price matches its historical mean level ($P=76$ €/m^3).

It follows that with a constant CO_2 price, regardless of its level, the presence of carbon trading leads to extending the optimal rotation period, when compared to the case without carbon sequestration. Furthermore, we observe that this effect is stronger with a higher CO_2 price. For example, for $\theta=1$, $\beta=1\%$ and $P^c=10$ €/tCO_2, the optimal rotation period increases from benchmark 80 years (the case without carbon sequestration) up to 86 years. Moreover, if $P^c=50$ €/tCO_2, the length of the optimal rotation period reaches even 155 years! However, the results are not that unexpected: since it is assumed that the CO_2 price stays constant as time passes, the forest owner knows with certainty how much he/she is obliged to pay when the harvest is performed. The higher the CO_2 price and also the higher θ, the longer it takes to override the negative carbon payment related to the harvest action.

For sensitivity of the results with respect to the discount rate, we conclude that regardless of the level of the CO_2 price the higher the discount rate, the shorter the optimal rotation period. This is similar to the case when we ignore carbon sequestration incentives.

However, the level of CO_2 price matters comparing to the effect of the carbon scheme setup: if the CO_2 price is assumed to stay on a relatively low level (10 €/tCO_2) the setup of the carbon scheme (that is parameter θ) does not induce any change on the optimal rotation length. On the other hand, if the CO_2 price is 50 €/tCO_2, the results are significantly sensitive to the setup of the carbon scheme parameter θ: changing θ from 0.3 to 1 results in approximately 30% increase of the optimal rotation period.

4.2.2. Constant wood price, exponentially increasing CO_2 price

Although in most papers on carbon sequestration the carbon and CO_2 price is kept constant, this assumption will most likely not coincide with the real world situation in the future. This is mainly due to the systematically increasing amount of greenhouse gases in the atmosphere in combination with the obligation of countries to meet environmental targets. As discussed in Section 3, the IIASA MESSAGE global carbon scenarios are also constructed within these boundaries. Therefore, we extend our study and analyze the problem under exponentially increasing CO_2 prices. Setting $\sigma^c=0$ in the GBM process (7) results in the required property. Again, the wood price remains constant.

A complete specification of the CO_2 price process requires us to fix the starting price. We therefore consider three values: 2, 5 and 10 €/tCO_2 as the starting CO_2 price. The results of the numerical computations are presented in Table 2.

Table 2
Optimal rotation period (in years) as a function of carbon scheme, discount rate and assuming a *constant* wood price (76 €/m^3) and *exponentially increasing* CO_2 price

Carbon scheme θ	Discount rate β	Optimal rotation $P^c=2$	Optimal rotation $P^c=5$	Optimal rotation $P^c=10$
1	1%	68	60	52
1	2%	58	55	51
1	3%	51	49	49
0.7	1%	72	65	61
0.7	2%	61	60	58
0.7	3%	52	52	54
0.3	1%	80	80	80
0.3	2%	65	68	72
0.3	3%	54	58	61

The starting values of CO_2 price are 2, 5 and 10 €/tCO_2, respectively.

[7] Note that the current trading price for carbon on the European Market is in excess of 20 €/tCO_2.

Table 3
Optimal rotation period (in years) with different model specifications

Wood price P	CO_2 price P^c	Carbon scheme θ	Discount rate β	Optional rotation (mean)	Optional rotation (volatility)
Constant	Constant	1	1%	82	–
Stochastic	Constant	1	1%	86.17	14.24
Constant	Increasing	1	1%	68	–
Stochastic	Increasing	1	1%	70.55	10.19
Constant	Stochastic	1	1%	71.54	7.39
Stochastic	Stochastic	1	1%	73.19	14.13

The starting price of wood is 76 €/m^3 and the starting value for the CO_2 price is 2 €/tCO_2.

The sensitivity analysis with respect to the discount rate leads to the same conclusion as observed in the previous two setups (that is, in the benchmark case and the constant CO_2 price case): the higher the discount rate, the shorter the optimal rotation period. Furthermore, the sensitivity of the results with respect to the carbon liability parameter θ can be expressed as follows: when θ decreases, the optimal rotation period becomes longer. Moreover, in neither this case nor the previous one, does the starting level of the CO_2 price affect the results.

The decrease in the optimal rotation length caused by higher θ can be explained as follows: the exponentially increasing CO_2 prices compete with the discounting effect and the probable future increase of the standing volume. In the case $\theta=1$, that is, when carbon credits must be purchased back for all of the previously sequestered carbon, the expectation of high costs due to released CO_2 reduces the length of the optimal rotation period. If θ were smaller, the expected trend of the rising CO_2 price would become less significant for the optimal rotation period length. A lower θ means that less carbon credits must be purchased at the harvesting stage, hence the increase in the CO_2 price is compensated by less carbon credits needed to be purchased.

Unlike the case of constant CO_2 price, the assumption of exponentially increasing CO_2 prices induces ambiguous effect on the length of the optimal rotation period in comparison to the benchmark case. A combination of the pair of parameters (θ, β) is crucial in the determination of the optimal rotation length. Under some setups (e.g., $\theta=1$ and $\beta=1\%$) the observed rotation period is shorter, for e.g., $\theta=0.3$ and $\beta=1\%$ the length of the optimal rotation period is approximately the same and in some cases the optimal rotation period is even longer (e.g., $\theta=0.3$ and $\beta=3\%$) compared to the benchmark case without carbon sequestration. These results suggest that for any given β it might be possible to set θ so that the resulting optimal rotation period is approximately the same length as without carbon sequestration.

4.3. Stochastic setup

Now we turn our attention to the stochastic case. In the following sections a detailed analysis of the optimal rotation period determination under uncertainty will be provided. We start by separately exploring the uncertainty effect of the wood price and the CO_2 price. Later in this section we present the results for the case when both prices are assumed to be stochastic, too.

4.3.1. Adding the uncertainty

In this section we study decision making under uncertainty. We proceed step-by-step; namely, we determine the optimal rotation period length for the following four different setups:

- Stochastic wood price, constant CO_2 price: $\mu^c=0$ and $\sigma^c=0$.
- Stochastic wood price, exponentially increasing CO_2 price: $\sigma^c=0$.
- Constant wood price, stochastic CO_2 price: $\alpha=0$ and $\sigma=0$.
- Stochastic wood price, stochastic CO_2 price.

The parameters that are not set to zero are considered to have the values estimated in Section 3. We compare the length of the optimal rotation period determined under all these stochastic setups with the corresponding

Table 4
Optimal rotation period (in years) with different model specifications

Wood price P	CO_2 price P^c	Carbon scheme θ	Discount rate β	Optional rotation (mean)	Optional rotation (volatility)
Constant	Constant	0.3	1%	82	–
Stochastic	Constant	0.3	1%	86.39	13.82
Constant	Increasing	0.3	1%	80	–
Constant	Stochastic	0.3	1%	80.83	0.62
Stochastic	Increasing	0.3	1%	85.58	13.26
Stochastic	Stochastic	0.3	1%	84.89	12.92

The starting price of wood is 76 €/m^3 and the starting value for the CO_2 price is 2 €/tCO_2.

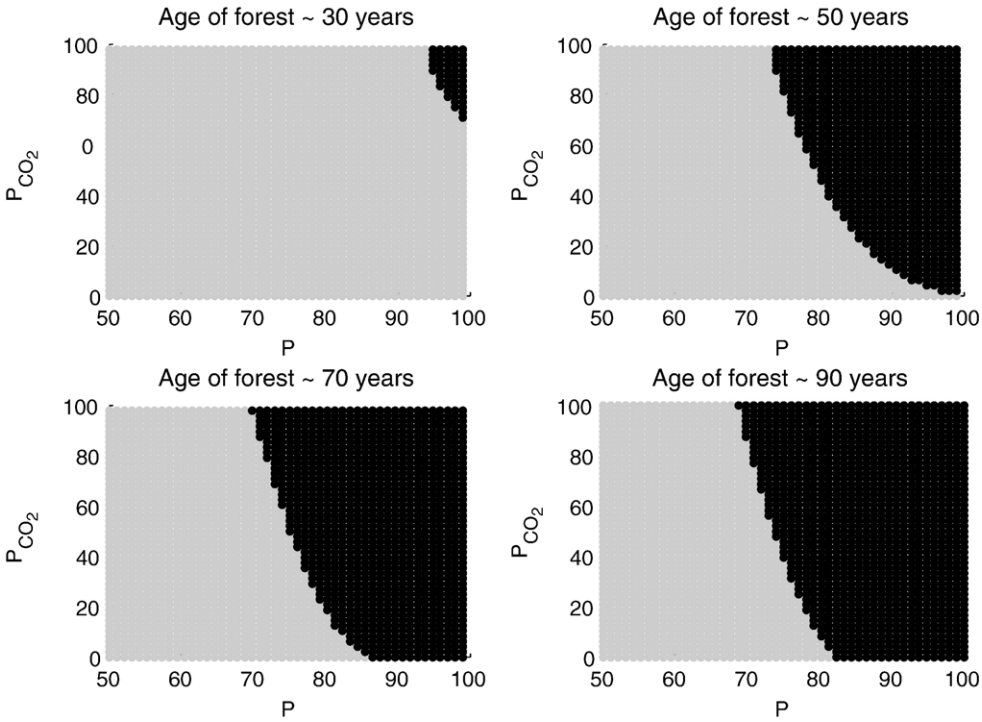

Fig. 3. Optimal decisions for a forest of age 30, 50, 70, and 90 years. The case of stochastic wood and stochastic CO_2 prices; discount rate β=1%, θ=1. The grey area represents the states for which it is optimal to postpone the harvest and the black area depicts those for which it is optimal to harvest immediately.

length gained using purely deterministic models presented in Section 4.2. Since the main aim is to explore the effect of the uncertainty, we fix the discount rate to 1% and consider only two carbon schemes: θ=1 and θ=0.3. Tables 3 and 4 summarize the results for all relevant models. For stochastic models, the mean value together with its standard deviation is presented.

In Table 3 the results are presented so that the impact of the wood price uncertainty can be observed. On the other hand, Table 4 groups the results stressing the effect of the CO_2 price uncertainty.

In all cases, the presence of uncertain wood prices leads to extending the optimal rotation period. An average optimal period is now approximately 2.4–5.6 years longer than the corresponding deterministic models predict. At the same time, a relatively high volatility in the results, due to stochastic wood prices, is observed.

A similar effect is apparent if uncertainty is added to the CO_2 price. However, the increase is less significant: the optimal rotation period is extended by a maximum of 3.5 years only. Moreover, the results show that under low θ (see Table 4), the stochasticity of the CO_2 prices makes almost no difference. A volatility induced by the stochastic CO_2 prices is also smaller compared to volatility induced by stochastic wood prices. Actually, including uncertainty in the CO_2 prices, the volatility of the results for θ=0.3 hardly changes.

A discrepancy between the impact of the stochastic wood and the stochastic CO_2 prices arises from the different proportion of their payoffs in the overall profit. Under the presented setup of parameters, selling wood is in relative terms more profitable than carbon sequestration. Therefore, it is more important to sell the wood at the "right" moment: once the wood growth starts to decline, the forest owner can decide to wait for a favorable situation — that is until the wood price temporarily increases. Since the wood price is assumed to follow a mean reverting process (1), waiting makes sense. This is especially true, when the wood price is low (i.e., below the average level): the mean reverting property assures that the wood price returns to at least its average level.

Table 5
Optimal rotation period (in years) with different model specifications

Wood price P	CO_2 price P^c	Carbon scheme θ	Discount rate β	Optimal rotation (mean)	Optimal rotation (volatility)
Stochastic	Stochastic	1	1%	75.08	14.39
Stochastic	Stochastic	1	2%	62.15	10.81
Stochastic	Stochastic	1	3%	53.43	8.32
Stochastic	Stochastic	0.7	1%	78.58	13.85
Stochastic	Stochastic	0.7	2%	63.99	10.70
Stochastic	Stochastic	0.7	3%	54.71	8.59
Stochastic	Stochastic	0.3	1%	84.89	12.92
Stochastic	Stochastic	0.3	2%	68.34	10.89
Stochastic	Stochastic	0.3	3%	57.29	9.34

The starting price of wood is 76 €/m^3 and the starting value for the CO_2 price is 2 €/tCO_2.

This "speculative" incentive is, however, not so strong when uncertainty is added to the CO_2 price. In this case, the overall shorter optimal rotation period and smaller variation is due to modelling the CO_2 price process as GBM (see Eq. (7)). Under this assumption, the CO_2 price will, on average, rise in future. This means that postponing the harvest and waiting until the CO_2 price temporarily decreases might be appropriate for a relatively short period only. Note, that the θ-level of the setup can influence the presence of the "speculative" incentive as well.

4.3.2. Stochastic wood price, stochastic CO_2 price

Finally, we present the results for the model with carbon sequestration with stochastic wood and stochastic CO_2 prices. Fig. 3 presents a typical shape of the decision diagrams under the stochastic prices.

The presented sequence of the decision diagrams documents the facts already pointed out in the previous sections: the main factor driving the forest management decision is the price of wood. Several general conclusions can be made on the basis of these figures:

- If the wood price is low,[8] irrespective of the CO_2 price level and the standing volume, the optimal decision is to postpone the harvest.
- Young forest stand always implies that the harvest will be postponed.
- If standing volume of the forest is close to its maximum and the wood price is close or higher than its mean value, it is optimal to perform the "harvest and sell" action.
- The higher the wood price, the less significant the role of the CO_2 price.

Finally, Table 5 summarizes the results obtained by performing numerical experiments under stochastic prices. In these experiments, the same distinguished values of the discount rate β and of the parameter θ have been chosen, as in the experiments with deterministic setups described in Section 4.2.

As we have already discussed, adding uncertainty to the prices extends the expected optimal rotation period comparing with the corresponding fully deterministic case analyzed in Section 4.2. The results show that waiting is more valuable under a low discount rate (i.e., β=1%). Moreover, in this case the highest variation of the expected optimal rotation is observed. This fully coincides with the explanation of the speculative owner stated in the previous section: the lower the time preference of the owner, the longer he/she is willing to wait for a payment and for a "more favorable" price situation and thus the optimal rotation will fluctuate more.

The sensitivity analysis of the carbon scheme setup shows the same trends as observed in the deterministic case (cf. Table 2). For more details see discussion in Section 4.2. We also obtain an identical conclusion for the discount rate: the higher the discount rate, the shorter optimal rotation period. In addition, discounting also influences the corresponding volatility of the revenue streams. The results show that that higher the discount rate, the smaller the volatility of the optimal rotation period length.

5. Conclusions

The model proposed in this paper extends the existing literature on the determination of optimal rotation periods. A real options analysis of the forest owner maximization problem is presented, which in addition to the revenue induced by selling wood also includes the payments arising from carbon sequestration. To our knowledge, this is probably the first paper

[8] All statements apply to achievable values of wood prices only, i.e. to the data in the range 50–100 €/m^3. This range corresponds to the historical observations.

providing a detailed numerical analysis using a real options framework considering both stochastic wood and stochastic CO_2 prices.

The numerical experiments presented demonstrate that the evolution of the prices has a strong impact on the optimal decision. We have discussed the case of increasing as well as constant CO_2 prices. Carbon price trajectories were obtained from a global integrated assessment model. We can conclude that only strong CO_2 price signals make a significant difference in terms of changes in the optimal rotation period.

In all presented model specifications a decrease in the discount rate leads to the extension of the optimal rotation period regardless of other parameters setup. Another important factor driving the optimal forest management decisions appeared to be the level of the forest owner's responsibility for the released CO_2 at harvest time. We have examined the optimal forest management problem under three carbon crediting schemes. We have shown that under the proposed carbon trading rules (except the case of the constant CO_2 price) the expected length of the optimal rotation period decreases as the forest owner's responsibility for the released CO_2 at the time of harvest increases.

The numerical results show that there is no definite statement about the impact of carbon trading on the length of the optimal rotation period. The optimal decisions are rather sensitive to many parameters and parameter constellations. The most significant extensions of the optimal rotation period compared to the case without carbon sequestration has been observed only under a high and constant CO_2 price. This evokes a conjecture that the forest owner is willing to prolong the rotation period and hence store the sequestered CO_2 for a longer time only if carbon sequestration is highly beneficial and payments are constant over time and certain. On the other hand, it is important to notice that under certain parameter setups carbon trading might induce that the optimal rotation period becomes *shorter* compared to the case with no carbon trading. The most significant relative shortening has been observed in case of a constant wood price and increasing CO_2 price, low discount rate ($\beta = 1\%$) and full liability to carbon stocks ($\theta = 1$). Under stochastic prices of carbon and timber this effect is less noticeable.

On the aggregate, the originally well-intended incentives to shorten rotation periods will actually lead to a reduction of carbon stocks in the forest. Such changes are in contradiction with the provisions of the Kyoto Protocol as well as the UNFCCC. Hence, carbon sequestration policies have to be carefully crafted in order to provide the right long-term incentives for forest owners. From our analysis, the institutional design with respect to reward and liability schemes appears to be of crucial importance.

Nonetheless, we have provided an extensive sensitivity analysis of the main factors that might induce shifting the length of the optimal rotation period. The presented uncertainty study, which overcomes the scope of traditional deterministic modelling, leads to new interesting considerations: the "speculative" incentive might be one of the factors why, under some circumstances, the expected optimal rotation period is extended. In the stochastic setting presented in this paper it appears that there is a value to wait for new information: the forest owner will try to benefit from temporarily higher prices for wood and lower prices for CO_2. In order to generalize this observation, it would be necessary to perform further numerical experiments in this regard with a wider variety of classes of stochastic processes. This, however, is beyond the scope of this paper.

The proposed real options model can be applied to a much broader class of problems than just determining the optimal rotation period. First, it enables derivation of the management rules for the forest in each phase of its growth — our algorithm is proposed in such a way that at each time t it calculates the optimal decision for each triple (P_t, P_t^c, S_t). Second, as indicated in Section 2, the set of handled management options can be expanded. The most natural extension is to include the option to perform thinning. Concerning the CO_2 mitigation issue it might also be of interest to elaborate on land use implications, e.g., an option to convert the forest area to agricultural land or to other commercial uses. Moreover, after some modifications it would be of interest to derive optimal management rules for a sequence rotations.

Variations to forest growth assumptions offers another interesting direction for future research. The impact of carbon sequestration with respect to the location (i.e. variations in the site index) or among different tree species (e.g., among fast and slow growth species) would direct our attention to the global consequences of carbon trading in the forestry sector.

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